

How I Built This Home SOC Lab

A Practical Implementation using Wazuh SIEM

1 Project Overview

I built a small Home Security Operations Center (SOC) lab using Oracle VirtualBox to gain practical, hands-on experience with centralized log monitoring, threat detection, and incident alerting. My primary goal was to set up a secure, isolated virtual environment where I could deploy the Wazuh SIEM platform, simulate cyber attacks, and observe how a SOC detects and responds to malicious activity in real-time.

2 Lab Setup and Architecture

To ensure my simulations wouldn't interfere with my home network, I built the lab on an isolated VirtualBox NAT network using the 192.168.100.0/24 subnet.

I configured static IP addresses for reliable communication and deployed three distinct virtual machines to represent a basic enterprise environment:

System Role	Operating System	Static IP Address
Wazuh Server Node	Linux Environment	192.168.100.10
Target System	Ubuntu Linux	192.168.100.20
Attacker System	Kali Linux	192.168.100.30

Table 1: Virtual Machine Network Allocation

3 Implementation Steps

3.1 Deploying the Infrastructure

First, I created the isolated VirtualBox network and deployed the three virtual machines. Once the networking was stable, I focused on the central monitoring server. I installed and configured the core Wazuh components: the Wazuh Manager (for centralized processing), the Wazuh Indexer (for log storage and search), and the Wazuh Dashboard (for visualization).

3.2 Configuring Endpoints and Services

Next, I installed the Wazuh Agent on the Ubuntu target machine and securely connected it to the Wazuh Manager to begin centralized log collection.

To make the lab functional for attack simulations, I intentionally enabled several services on the target system:

- **SSH (Secure Shell)** to test remote access and brute-force detection.
- **FTP (File Transfer Protocol)** to monitor cleartext credential transmissions.
- **Apache Web Server** to generate web traffic and access logs.

I also activated advanced endpoint monitoring capabilities within Wazuh, including File Integrity Monitoring (FIM), anomaly-based log analysis, and rootcheck mechanisms.

4 Attack Simulation and Detection

With the monitoring infrastructure in place, I switched to the Kali Linux attacker machine to validate the SIEM deployment. I performed several basic attack simulations against the Ubuntu target:

- Conducted aggressive port scanning to map out the target's available services.
- Launched automated SSH brute-force attempts targeting user credentials.
- Executed FTP login tests to generate authentication failure logs.

Results: The simulations successfully triggered alerts in the Wazuh Dashboard. I was able to see the raw logs, observe the rule-based event correlation, and review the high-severity alerts. This confirmed the SIEM was actively identifying suspicious behavior.

5 Project Outcome

Building this lab gave me practical, hands-on experience with SIEM deployment, endpoint monitoring, and basic incident detection. Beyond the security concepts, it also sharpened my administrative skills in Linux, network configuration, and virtualized infrastructure.